



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
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Lab 6

Combinational Circuits III

Contents

1	Objectives	2
2	Tasks	3
2.1	Easy Sort	3
2.1.1	Instructions	3
2.2	Mini ALU	4
2.2.1	Instructions	4
3	Instructions:	5

1 Objectives

The objectives of this experiment are as follows:

- Perform 2-digit BCD addition with proper correction and display.
- Compare two BCD digits using basic logic gates.
- Use BCD to 7-segment decoder (e.g., 4511) to drive displays.

2 Tasks

Today's lab task are as follows -

2.1 Easy Sort

In this task, design a combinational circuit that sorts three 4-bit binary numbers and displays the **largest**, **middle**, and **smallest** numbers.

2.1.1 Instructions

1. Use three 4-bit binary numbers (**A**, **B**, and **C**) as inputs.
2. Compare the input numbers using only two 4-bit comparator ICs (4063) and basic logic gates.
3. Use quadruple 2-to-1 multiplexers (74HC157 or 74LS157) to select between only two numbers at a time based on the comparator outputs.
4. Arrange the inputs in sorted order to determine the **Largest**, **Middle**, and **Smallest** numbers.
5. Display the sorted outputs using suitable 7-segment display modules.
6. Verify the circuit by testing different input combinations to ensure the outputs are correctly sorted for all possible cases.

Allowed Components

- Basic logic gates (AND, OR, NOT, XOR, etc.)
- 4-bit comparator ICs (4063)
- Quadruple 2-to-1 multiplexers (74HC157 or 74LS157)
- Input switches or 4-bit binary input sources
- 7-segment display modules (7SEG-BDC)

Testing

Verify the circuit by applying different permutations of the same three input values. For example, use the values **2 (0010)**, **9 (1001)**, and **5 (0101)** in different input orders, such as:

$$(A, B, C) = (2, 9, 5), (2, 5, 9), (9, 2, 5), (9, 5, 2), (5, 2, 9), (5, 9, 2).$$

For every permutation, the outputs must remain:

$$\mathbf{Largest} = 9 \text{ (1001)}, \quad \mathbf{Middle} = 5 \text{ (0101)}, \quad \mathbf{Smallest} = 2 \text{ (0010)}.$$

Repeat the test with several other sets of three 4-bit numbers to verify correct operation.

2.2 Mini ALU

In this task, design and implement a simple combinational Arithmetic Logic Unit (ALU) that performs different logical operations on two 3-bit binary numbers based on a 3-bit control input.

2.2.1 Instructions

1. Use two 3-bit binary numbers (**A** and **B**) as inputs.
2. Use control inputs (**S**) to select one of the ALU operations.
3. Implement the following operations:

S	Operation
00	Output the parity bit of A
01	Output the parity bit of B
10	Output 0 if $A > B$; otherwise output 1 .
11	Output 1 if $A = B$; otherwise output 0 .

4. Display the 1-bit output using suitable output indicators or display modules.
5. Verify the circuit by testing all operations with different combinations of input values.

Allowed Components

- Basic logic gates (AND, OR, NOT, XOR, etc.)
- Multiplexers (74HC157 or 74LS157)
- Input switches
- LEDs or 7-segment display modules for output

Testing

Test the ALU by applying different values of **A**, **B**, and the control input **S**. Verify that each control code performs the correct operation. For example, when $A = 101$ and $B = 101$, selecting operation **11** should produce an output of **0**, indicating that the two inputs are equal.

3 Instructions:

1. **File Naming Format:** ID_L06T01 and ID_L06T02.
2. For task 2 you have to come up with the explanation of how you are showing the result in a seven segment display.
3. You will get 1 hour to complete these tasks.
4. The rubrics for today's lab task can be found [here](#).